Zhaofeng Sun

sun-zf22@mails.tsinghua.edu.cn / Website / Google Scholar / GitHub

Research interests: Language model architectures, machine learning systems, and model compression.

Education

Tsinghua University Sep 2022 - Jun 2027 (expected)
B.S. in Computer Science | GPA: 3.97/4.00 | Rank: 5/208

Cornell University Jan 2025 - Jun 2025
Exchange Student in Computer Science | GPA: 4.15/4.00

Research Experience

University of California San Diego Jun 2026 - Present
Prof. Hao Zhang · Hao AI Lab

- **ForesightLoop.** Makes additional recurrent computation effective in parallel language models, improving training stability and depth scaling. At eight loops, improves average downstream accuracy by 0.84 percentage points over vanilla PLT with only 0.3% additional decoding time.

Carnegie Mellon University May 2025 - Aug 2025
Prof. Beidi Chen

- **Sparrow.** Makes long-context reinforcement learning more efficient by controlling the mismatch between sparse rollout actors and dense training policies, sustaining stable learning with up to 2.4× faster rollouts.

Cornell University Jan 2025 - May 2025
Prof. Christopher De Sa

- **Model-Preserving Adaptive Rounding.** Preserves model behavior after low-bit quantization by targeting end-to-end output error, with theoretical guarantees and approximately 30% lower KL divergence than GPTQ/LDLQ.

Tsinghua University Oct 2023 - Jan 2025
Prof. Jianfei Chen

- **Efficient model compression.** Hardware-aware sparsity and quantization connect compression choices to accelerator execution, aiming to turn model compression into practical inference speedups.

Selected Publications

Model-Preserving Adaptive Rounding
Albert Tseng, **Zhaofeng Sun**, Christopher De Sa.
ICML 2026 Paper

Sparrow: Sparse Rollout for Stable and Efficient Long-context RL of Large Language Models
Yang Zhou, Ranajoy Sadhukhan, **Zhaofeng Sun**, Zhuoming Chen, Souvik Kundu, Saket Dingliwal, Sai Muralidhar Jayanthi, Aram Galstyan, Haizhong Zheng, Beidi Chen.
Preprint, 2026 · Earlier version at SPOT, ICLR 2026 Paper

Beyond 2:4: Exploring V:N:M Sparsity for Efficient Transformer Inference on GPUs
Kang Zhao, Tao Yuan, Han Bao, Zhenfeng Su, Chang Gao, **Zhaofeng Sun**, Zichen Liang, Liping Jing, Jianfei Chen.
Preprint, 2024 Paper

Honors and Technical Skills

Champion, Tsinghua University Supercomputer Competition (2024).

Comprehensive Excellence Scholarship, top 5%, Tsinghua University (2023-2025).

Languages: Python, C/C++, CUDA, Triton. **Systems:** PyTorch, FSDP, Megatron, vLLM, SGLang, verl.